

Autosomal Recessive Nonsyndromic Hearing Loss 48 (DFNB48 / CIB2-related deafness)

Comprehensive Disease Characterization Report

Summary (Answer to the Research Question)

Autosomal Recessive Nonsyndromic Hearing Loss 48 (DFNB48; OMIM #609439) is a monogenic, autosomal-recessive, **prelingual, bilateral, symmetric, generally severe-to-profound sensorineural hearing loss** caused by **biallelic loss-of-function or missense variants in *CIB2*** (calcium- and integrin-binding protein 2; OMIM *605564, HGNC:24579, 15q25.1). *CIB2* is an **essential auxiliary subunit of the hair-cell mechanoelectrical transduction (MET) channel complex**, binding TMC1/TMC2 at the tips of cochlear stereocilia and, independently, organizing the stereocilia "staircase" via whirlin (WHRN). Its loss abolishes cochlear MET despite intact tip links, causing hair-cell dysfunction and death and thus deafness. The phenotype is **nonsyndromic** — vision and balance are spared because the paralog ***CIB3*** compensates in the vestibule and retina — a point that revises the earlier assignment of *CIB2* to Usher syndrome type 1J. Management is currently rehabilitative (hearing aids, **cochlear implants**), but AAV-mediated *CIB2/CIB3* gene therapy rescues hearing in mouse models, and clinically validated cochlear gene therapy for the analogous *OTOF/DFNB9* deafness signals a plausible future disease-modifying route.

Evidence base: human consanguineous pedigree/cohort studies, engineered mouse (knockout/knock-in), zebrafish and *Drosophila* models, and in-vitro biochemistry/structural modeling.

1. Disease Information

- **Overview:** A rare, inherited form of nonsyndromic sensorineural hearing impairment presenting at/before birth, caused by dysfunction of the cochlear hair-cell mechanotransduction apparatus. It is isolated (nonsyndromic) — hearing loss is the only clinical feature.
- **Key identifiers:**
- **OMIM (phenotype):** #609439 — "Deafness, autosomal recessive 48 (DFNB48)"
- **Gene:** *CIB2*, OMIM *605564; **HGNC:**24579; **NCBI Gene:** 10518; **Ensembl:** ENSG00000136010; **UniProt:** O75838; locus **15q25.1**
- **MONDO:** maps to OMIM:609439 (autosomal recessive nonsyndromic hearing loss 48)
- **ICD-11:** AB52 (sensorineural hearing loss) / hereditary hearing impairment; **ICD-10:** H90.3–H90.5 (no DFNB48-specific code)
- **MeSH:** "Hearing Loss, Sensorineural"; "Deafness"
- **Orphanet:** within "Rare genetic deafness / autosomal recessive nonsyndromic sensorineural deafness type DFNB"
- **Synonyms:** DFNB48; Deafness, autosomal recessive 48; *CIB2*-related nonsyndromic hearing loss.
- **Information source:** Aggregated disease-level resources (OMIM/Orphanet) plus individual patient/consanguineous-pedigree studies (predominantly Pakistani, plus Iranian, Turkish, Dutch families). Evidence type: human clinical + model organism + in vitro.

Primary

citations:

P 23023331

(Riazuddin 2012, discovery);

P 29112224

(Booth 2018, NSHL vs USH).

2. Etiology

- **Primary cause — genetic:** Biallelic pathogenic variants in *CIB2* (necessary and sufficient). Loss-of-function (nonsense, frameshift, splice) and specific missense variants cause disease

via **loss of function**; heterozygous carriers are unaffected

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- **Genetic risk factors:** The *CIB2* genotype itself; **consanguinity / autozygosity** (94.4% of affected individuals in Pakistani cohorts are homozygous;

P 30303587);

founder ancestry carrying **c.272T>C p.(Phe91Ser)**.

- **Environmental risk factors: None established.** DFNB48 is not caused by toxins, noise, occupational exposure, radiation, diet, or lifestyle. General ototoxins (aminoglycosides) and noise are not specifically linked to *CIB2* (contrast *MT-RNR1* m.1555A>G). Sex: no predilection (autosomal). Family history/consanguinity are the operative demographic risks.

- **Protective factors:** No human protective alleles identified. Biologically, the paralog **CIB3** provides intrinsic redundancy in the vestibule and retina (limiting the phenotype to hearing) but does **not** protect the cochlea

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P 34089643
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- **Gene–environment interactions:** None documented specifically for *CIB2*.
- **Epigenetic / chromosomal mechanisms:** None known; no recurrent CNV/structural or methylation mechanism reported.

3. Phenotypes

Core phenotype (100% of affected): bilateral sensorineural hearing loss.

Phenotype	HPO term	Characteristics
Sensorineural hearing impairment	HP:0000407	Clinical sign; the defining feature
Bilateral SNHL	HP:0008619	Bilateral, symmetric
Congenital/prelingual SNHL	HP:0008527 / HP:0008573	Onset at birth / before speech
Profound SNHL	HP:0011476	Most common severity (esp. LOF genotypes)
Severe SNHL	HP:0008625	Also reported; some milder cases

- **Onset:** neonatal/congenital or prelingual.
- **Severity:** typically **severe-to-profound**; **variable** — deafness with c.272T>C in one Pakistani family was "remarkably less severe" than in other families with the same allele, indicating variable expressivity/modifiers
([P 26173970](#)).
- **Progression:** generally **stable/non-progressive**; chronic, lifelong.
- **Frequency among affected:** hearing loss ~100%; **notably ABSENT** are retinitis pigmentosa (HP:0000510) and vestibular dysfunction — their absence defines the nonsyndromic status vs USH1J.
- **Quality-of-life impact:** primary burden is on **spoken-language acquisition, communication, education, and psychosocial development**; largely mitigated by early amplification/cochlear implantation. No pain, no systemic morbidity.

Citations:

[P 29112224](#)

[P 26173970](#)

[P 29086887](#)

4. Genetic / Molecular Information

- **Causal gene:** *CIB2* — calcium- and integrin-binding family member 2 (HGNC:24579; OMIM *605564; NCBI 10518; UniProt O75838). Encodes a small (~187 aa) EF-hand protein with calcium-binding domains.

- **Pathogenic variants (representative):**

- **c.272T>C, p.(Phe91Ser)** — recurrent **founder missense**, prevalent DFNB48 cause in Pakistan

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- **c.196C>T p.(Arg66Trp), c.97C>T, c.556C>T** and other missense

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- **Nonsense/LOF:** e.g., p.(Gln12), p.(Tyr110)

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 frameshift and additional LOF alleles
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- **Variant classification:** Pathogenic / likely pathogenic per ACMG/AMP (ClinVar). Biallelic LOF → ARNSHL (not USH)

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- **Variant types:** missense, nonsense, frameshift, splice-site (predominantly missense and LOF); no recurrent structural/CNV mechanism.

- **Allele frequency:** individual pathogenic alleles are rare in gnomAD; founder alleles enriched in South-Asian subpopulations.

- **Origin:** **germline** (constitutional); not somatic.

- **Functional consequence: loss of function** — disrupted CIB2–TMC1/TMC2 interaction and impaired stereocilia organization; missense alleles are proposed to alter integrin binding and channel modulation while preserving localization/calcium buffering in some cases

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- **Modifier genes:** *CIB3* (paralog, functional redundancy); *WHRN* (genetic interaction in mice; P 40083274).
- Other MET-complex genes (*TMC1/2*, *LHFPL5*, *TMIE*, *LOXHD1*) are functional partners.
- **Epigenetics / chromosomal abnormalities:** none established.

5. Environmental Information

Not applicable — DFNB48 is a **purely genetic, monogenic** disorder. No environmental toxin, radiation/pollution, occupational exposure, lifestyle factor (smoking/diet/alcohol/exercise), or **infectious agent** causes or triggers it. (Acquired congenital SNHL from e.g. congenital CMV/TORCH is a *differential diagnosis*, not a cause of DFNB48.)

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation)

1. **Biallelic pathogenic *CIB2* variant → loss of functional *CIB2* protein** (or a *CIB2* that cannot bind its partners). *[demonstrated]*
2. Loss of *CIB2* → **disrupted *CIB2*–*TMC1*/*TMC2* interaction** at the tips of shorter-row stereocilia → the **MET channel complex fails to operate**, even though tip links remain intact. *[demonstrated]* —

P 28663585

3. In parallel branch: loss of *CIB2* → **loss of *CIB2*–*WHRN*-dependent organization** of the stereocilia staircase → **abnormal stereocilia bundle morphology / overgrowth of shorter transducing-row stereocilia**. *[demonstrated]* —

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P 40083274

4. Non-functional MET → **failure to convert sound-induced mechanical deflection into receptor (transduction) current** → loss of hair-cell depolarization/signaling.

[demonstrated — abolished MET in mice]

5. Loss of transduction + CIB2's role in survival → **hair-cell dysfunction and progressive hair-cell degeneration** in the organ of Corti. *[demonstrated/inferred —*

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6. Cochlear sensory failure → **no afferent auditory signal to the cochlear nerve/CNS** → **congenital sensorineural hearing loss**. *[demonstrated]*

7. **Branch (spared organs):** In vestibule and retina, the paralog **CIB3** (and CIB1) substitutes → **balance and vision preserved** → phenotype remains **nonsyndromic**. *[demonstrated —*

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Detail by category

- **Molecular pathway / biochemical defect:** hair-cell **mechanotransduction** (an **ion-channel/receptor-complex defect**, not a classical signaling cascade). CIB2 is a **Ca²⁺-binding auxiliary subunit** of the TMC1/TMC2 pore complex (with TMIE, LHFPL5, PCDH15). EF-hand calcium binding modulates the complex.
- **Protein dysfunction:** loss of function; disrupted protein–protein interactions (TMC1/2, WHRN). Missense variants may impair integrin/partner binding.
- **Cellular processes:** impaired sensory transduction; **hair-cell degeneration/death**; dysregulated **stereocilia actin cytoskeleton/bundle morphogenesis**.
- **Metabolic / immune / inflammatory:** not implicated.
- **Tissue damage mechanism:** sensory hair-cell loss (not fibrosis/ischemia/autoimmune).
- **GO terms:** sensory perception of sound (GO:0007605); inner ear receptor cell development (GO:0060113); regulation of stereocilium/microvillus length; calcium ion binding (GO:0005509). **Cellular component:** stereocilium (GO:0032420), stereocilium tip (GO:0032426), stereocilium bundle (GO:0032421).
- **Cell types (CL):** cochlear inner hair cell (CL:0000589), cochlear outer hair cell (CL:0000601), auditory hair cell (CL:0000202).

Citations:

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P 34089643

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7. Anatomical Structures Affected

- **Organ level (primary): cochlea** (UBERON:0001844), specifically the **organ of Corti / spiral organ** (UBERON:0002227) within the **inner ear** (UBERON:0001846). **Body system:** nervous/sensory (auditory).
- **Secondary involvement:** none directly; downstream central auditory pathway deprivation affects language cortex development if untreated.
- **Tissue/cell level:** auditory **sensory epithelium (neuroepithelium)**; affected cells = **cochlear inner and outer hair cells** (CL:0000589 / CL:0000601). Vestibular hair cells and retinal photoreceptors are **spared** in humans.
- **Subcellular level:** the **stereocilia** and their **tips** (mechanotransduction apparatus); GO:0032420 stereocilium, GO:0032426 stereocilium tip; plasma membrane; EF-hand Ca²⁺-binding protein.
- **Localization / lateralization:** **bilateral, symmetric** (HP:0008619).

Citations:

P 23023331

(stereocilia localization); 28663585.

8. Temporal Development

- **Onset: congenital / prelingual;** onset pattern is **chronic** (present from birth), not acute or episodic.
 - **Progression:** generally **stable / non-progressive; variable severity** across genotypes/families. Disease **duration is lifelong (chronic).**
 - **Course pattern:** persistent; no relapsing-remitting or fluctuating course; **no spontaneous remission.**
 - **Critical period: early childhood auditory neuroplasticity** window — early diagnosis and habilitation (EHDI 1-3-6 benchmark: screen ≤ 1 mo, diagnose ≤ 3 mo, intervene ≤ 6 mo) determine language outcomes; preclinical data show a developmental window for gene-therapy rescue
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9. Inheritance and Population

- **Inheritance: autosomal recessive** (biallelic). Recurrence risk **25%** per pregnancy for carrier couples.
 - **Penetrance: complete** for biallelic pathogenic genotypes. **Expressivity:** variable (severity differs even for identical alleles;
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- **Anticipation / mosaicism:** not applicable / not reported.
 - **Founder effects: c.272T>C p.(Phe91Ser)** is a prevalent Pakistani founder allele
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- **Consanguinity:** major driver (homozygosity by descent;
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- **Carrier frequency:** rare in outbred populations; elevated in specific consanguineous kindreds; individual alleles rare in gnomAD.

- **Epidemiology:** Overall congenital SNHL prevalence $\approx$ **1–3 per 1,000** newborns; DFNB48 is a **small fraction** of this globally but is one of the more common ARNSHL genes in Pakistan — **4th most common** after *SLC26A4*, *MYO7A*, *GJB2*

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and among 13 genes accounting for >50% of profound HL in Pakistan

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No precise DFNB48 prevalence/incidence figure is established (rare disease).

- **Demographics:** enriched in **South Asian (Pakistani/Iranian)**, **Turkish**, and some **European (Dutch)** kindreds; **no sex bias**; affected from birth (pediatric age distribution at diagnosis).

10. Diagnostics

- **Audiological (clinical) tests:**

- **Screening:** Universal Newborn Hearing Screening — **otoacoustic emissions (OAE)** + **automated ABR (AABR/BERA)**

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- **Confirmatory:** diagnostic **ABR**, **pure-tone/behavioral audiometry**, tympanometry (to exclude conductive loss). LOINC-coded audiometry/OAE/ABR panels.
- **Imaging:** temporal-bone **CT/MRI** to exclude structural inner-ear malformations (usually normal in DFNB48).
- **Laboratory biomarkers:** **none** — no blood/urine/enzyme/metabolic biomarker; diagnosis is audiological + molecular.
- **Genetic testing (definitive):**
- **NGS gene panels / whole-exome sequencing** targeting 150–250+ deafness genes (including *CIB2*): diagnostic yield **~21–48%** for congenital SNHL

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[48%], 29986705 [42%], 35580552 [21%]).

- **Single-gene / targeted testing** appropriate in **founder populations** (e.g., tetra-primer ARMS for p.Phe91Ser;

P 29086887

- **WGS** for panel/exome-negative cases; **CMA/karyotype/FISH/mtDNA/repeat testing** are **not** indicated for isolated *CIB2* deafness.
- Variant interpretation per **ACMG/AMP** (ClinVar/ClinGen).
- **Clinical criteria / differential diagnosis:** bilateral nonsyndromic SNHL with normal vision and balance. **Rule out:** other ARNSHL (*GJB2/DFNB1*, *SLC26A4/Pendred*, *MYO15A*, *TMC1*, *OTOF/DFNB9*); **Usher syndrome (USH1J)** — assess vision/vestibular function; acquired causes (**congenital CMV/TORCH**).
- **Screening (asymptomatic):** newborn hearing screening; **carrier screening** and **cascade testing** of relatives.

11. Outcome / Prognosis

- **Survival/mortality: not life-limiting;** normal life expectancy; no disease-specific mortality.
- **Morbidity/disability: communication and language disability;** downstream educational and psychosocial impacts if unaddressed. ICF domains: hearing functions, communication, social participation.
- **Quality-of-life tools:** generic (EQ-5D, PROMIS) and hearing-specific instruments; outcomes strongly modifiable by intervention.
- **Disease course/complications:** stable auditory deficit; main "complication" is speech-language delay without early habilitation.
- **Recovery potential:** none spontaneously; **substantial functional recovery of hearing/oral language with cochlear implantation**, especially when performed early.
- **Prognostic factors:** **age at intervention** (earlier = better), rehabilitation intensity/appointment adherence, neurodevelopmental comorbidities, and cochlear/cochlear-nerve integrity on imaging

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No molecular prognostic biomarker beyond genotype–severity trends.

12. Treatment

- **Pharmacotherapy:** **None** corrects the MET defect (no approved drug; no pharmacogenomic modifier). NCIT: n/a.
- **Rehabilitative / device (standard of care):**
- **Hearing aids** (residual hearing) — NCIT: Hearing Aid.
- **Cochlear implantation** for severe-to-profound loss — NCIT: **Cochlear Implant**; paired with **auditory-verbal / speech-language therapy** (NCIT: Speech Therapy). Language outcomes depend on early implantation and rehabilitation
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- **Advanced/experimental — gene therapy:**
- **Preclinical (CIB2-specific):** single semicircular-canal **AAV-*Cib2*** (and **AAV-*Cib3***) injection restored stereocilia architecture and hearing in *Cib2*-mutant mice within a critical window
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- NCIT: Gene Therapy; AAV Vector.
- **Clinical precedent (analogous gene):** **AAV cochlear gene therapy for biallelic *OTOF*/DFNB9** — FDA-approved **Otarmeni** (2026); ~75% of early-trial children met the primary hearing endpoint at 24 weeks
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- ≥5 DFNB9 trials ongoing
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- Establishes feasibility; **CIB2 gene therapy remains preclinical.**
- **Surgical:** cochlear implant surgery (transmastoid/round-window).
- **Treatment strategy:** etiology-guided, early habilitation; emerging "**restore when biology permits, bypass when it does not**" hierarchy
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- Personalized medicine: genotype-directed eligibility for future gene therapy.

13. Prevention

- **Primary prevention:** no lifestyle/vaccine route (monogenic). Risk reduction is genetic: **preconception genetic counseling**, **expanded carrier screening**, and reproductive options — **prenatal diagnosis** and **preimplantation genetic testing (PGT-M)** for known familial variants; counseling on **consanguinity** risk. NCIT: Genetic Counseling; Carrier Screening; Prenatal Diagnosis; Preimplantation Genetic Testing.
 - **Secondary prevention: universal newborn hearing screening** → early diagnosis → early habilitation (EHDI 1-3-6).
 - **Tertiary prevention:** cochlear implantation + intensive (re)habilitation to prevent language/developmental morbidity; educational support.
 - **Cascade screening:** test at-risk relatives once a familial genotype is known.
 - **Public health:** deafness gene carrier-screening programs (already impactful for *GJB2* in some populations); consanguinity education in high-risk communities.
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14. Other Species / Natural Disease

- **Taxonomy / orthologs:** *CIB2* is evolutionarily conserved. Orthologs: **mouse *Cib2*** (NCBI Gene 208768; NCBI Taxon 10090), **zebrafish *cib2*** (Taxon 7955), ***Drosophila melanogaster*** ortholog (Taxon 7227) — all required for hair-cell/mechanosensory function (P 23023331).
 - **Natural disease / veterinary:** No spontaneous naturally occurring *CIB2* deafness is catalogued in OMIA as a recognized companion-animal/wildlife disease; disease models are **engineered**. (Not applicable / none reported.)
 - **Comparative pathology:** mouse and zebrafish reproduce hair-cell mechanotransduction failure; conservation of the MET-complex mechanism across vertebrates and invertebrates supports deep evolutionary conservation of *CIB2* function.
 - **Transmission:** not applicable (non-infectious, non-zoonotic).
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15. Model Organisms

- **Mouse (*Mus musculus*, primary model):**
- ***Cib2*-knockout** and **human deafness knock-in** lines: **deaf, no cochlear MET**, stereocilia bundle defects/overgrowth
([P 28663585](#));
([P 29255404](#))
- **Genetic-interaction model:** *Cib2*; *Whrn* double mutants
([P 40083274](#))
- **Recapitulation:** faithfully models profound congenital deafness and MET loss. **Limitation / species difference:** mice show **vestibular dysfunction** (vestibular CIB3 compensation differs from human) and **no retinal degeneration**, so mice do not fully mirror the human "hearing-only, balance-normal" phenotype.
- Resources: MGI, IMPC.
- **Zebrafish (*Danio rerio*):** *cib2* required for hair-cell function/development
([P 23023331](#));
resource: ZFIN.
- **Drosophila melanogaster:** CIB2 ortholog essential for hair-cell/mechanosensory development
([P 23023331](#));
resource: FlyBase.
- **In vitro / cellular:** heterologous expression (COS-7) for calcium-response and localization assays
([P 26173970](#));
AlphaFold2-multimer structural modeling of CIB2-WHRN
([P 40083274](#))
- **Applications:** dissecting MET-complex assembly, stereocilia morphogenesis, genotype–function relationships, and **AAV gene-therapy proof-of-concept**
([P 42427029](#))

Supported vs Refuted Hypotheses

Supported - *CIB2* biallelic variants cause DFNB48 via loss of hair-cell MET

([P 28663585](#)

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[P 29255404](#)

- Biallelic LOF causes **nonsyndromic** deafness, **not** Usher syndrome

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- *CIB2* has an independent WHRN-dependent role in stereocilia architecture

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- *CIB3* redundancy explains sparing of vestibule/retina

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- AAV gene therapy can rescue the cochlear phenotype in models

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clinically validated for the analogous OTOF deafness

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Refuted / revised - The original proposal that *CIB2* causes **Usher syndrome type 1J** is **refuted** for biallelic LOF genotypes

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CIB2 has been "disqualified as an USH-causing gene."

Limitations and Future Directions

- **Limitations:** No large natural-history registry or precise DFNB48 prevalence; genotype–phenotype (severity) correlations are incompletely explained (modifiers such as *CIB3* dosage remain hypothetical); most mechanistic evidence is from mouse/zebrafish, which imperfectly mirror the human sensory-organ-specificity; no human-specific molecular biomarker.

- **Future directions:** (1) *CIB2*-targeted AAV gene therapy toward first-in-human trials; (2) high-resolution structures of the *CIB2*–*TMC1* MET complex to guide variant interpretation and therapy; (3) systematic ACMG re-classification of *CIB2* VUS with functional assays; (4) modifier-gene (*CIB3*/*WHRN*) studies to explain variable expressivity; (5) equitable carrier-screening and counseling in high-consanguinity populations.
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Key References (PMID)

23023331 · 29112224 · 28663585 · 29255404 · 29084757 · 40083274 · 34089643 · 42427029 · 30303587 · 29086887 · 26173970 · 38534090 · 38224868 · 29986705 · 35580552 · 42181374 · 42470357 · 41812306 · 42330611 · 41895171

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